Sophia N. Wilson

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Biography

Sophia N. Wilson is a Ph.D. fellow at the Sustainable Artificial Intelligence for Sciences (SAINTS) Lab, Dept. of Computer Science, University of Copenhagen (UCPH), supervised by Raghavendra Selvan and Erik Dam. Her research spans two strands: a technical strand developing algorithms and AI systems that are more resource efficient in data, compute, and energy use; and a socio-technical strand measuring and mitigating sustainability harms in the AI industry, including environmental impact and poor labour conditions. She is also active in outreach activities promoting STEM fields, particularly to girls and young women.

Research Experience

Oct. 2026 –present **Visiting Researcher**, Samsung Semiconductor Denmark Research.
Research topic: Profiling the Hardware Utilisation of Agentic AI Workloads on Consumer Hardware.
Supervision: Vivek Shah.
Funding: AI Denmark.

Education

2025–present **PhD Fellow, Sustainable AI**, SAINTS Lab, Dept. of Computer Science, UCPH, Denmark.
Thesis: Glean: Towards Green Lean Data in the Era of Big Dark Data.
Supervision: Raghavendra Selvan and Erik Dam.
Funding: The Independt Research Fund Denmark.

2023–2025 **MSc in Computational Physics**, Niels Bohr Institute, UCPH, Denmark.
Thesis: "Quantifying the Reduction in Carbon Footprint of Physics-Informed Machine Learning".
Supervision: Raghavendra Selvan and Jens Hesselbjerg Christensen.

2019–2023 **BSc in Physics**, Niels Bohr Institute, UCPH, Denmark.
Thesis: "Measuring Cosmological Parameters with Fast Radio Bursts".
Supervision: Kasper Heintz and Johan Fynbo.

Selected Publications

In Conference Proceedings

2026 **Sophia N. Wilson**, Andrew Millard, Guðrún Fjóla Guðmundsdóttir, Raghavendra Selvan, and Sebastian Mair. Position: Stop Preaching and Start Practising Data Frugality for Responsible Development of AI. In *Proceedings of the 43rd International Conference on Machine Learning (ICML), Position Paper Track*, 2026.

2026 **Sophia N. Wilson**, Sebastian Mair, Mophat Okinyi, Erik B Dam, Janin Koch, and Raghavendra Selvan. How Hyper-Datafication Impacts the Sustainability Costs in Frontier AI. In *Proceedings of the 2026 ACM Conference on Fairness, Accountability, and Transparency (FAccT)*, 2026.

2026 **Sophia N. Wilson**, Jens Hesselbjerg Christensen, and Raghavendra Selvan. Performance-Carbon Trade-Offs across Architectural Biases in Shear Flow Forecasting. In *Proceedings of the 19th European Conference on Computer Vision (ECCV), LIMIT Workshop*, 2026.

2026 Pedram Bakhtiarifard, **Sophia N. Wilson**, Mahmoud Afifi, Jonathan Wenshøj, and Raghavendra Selvan. Characterizing Learning in Deep Neural Networks using Tractable Algorithmic Complexity Analysis. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2026.

Awards

Oct. 2026 **Best Poster (jury)**, Danish Digitalization, Data Science and AI (D3A) 4.0 Conference.

Aug. 2026 **Best Poster (public vote)**, Eastern European Machine Learning Summer School 2026, Cetinje, Montenegro.

Grants & Funding

2026 **Samsung/UCPH Knowledge Collaboration**, *AI Denmark*, 200,000 DKK, Co-applicant.

2024 **MSc thesis support**, *Stefan Rozental og Hanna Kobylinski Rozentals Fond*, 20,000 DKK.

2023–24 **Exchange semester at the University of Amsterdam**, *Seven funders*, 93,000 DKK in total, Stefan Rozental og Hanna Kobylinski Rozentals Fond (30,000 DKK), Erasmus+ (20,500 DKK), Knud Højgaards Fond (12,000 DKK), Nordea-fonden (10,000 DKK), William Demant Fonden (10,000 DKK), Niels Bohr Fonden (5,500 DKK), Dansk Kvindesamfundslegater (5,000 DKK)..

Teaching

2026 **Data Analysis and Machine Learning**, *lecturer and exam coordinator*, MSc course for high school teachers, Dept. of Computer Science, Aarhus University.

2026 **University College Copenhagen**, *guest lecturer*, BSc course on Ethics and Technology, "AI, Resource Use, and Sustainability".

2025 **Machine Learning A**, *teaching assistant*, MSc course, Dept. of Computer Science, UCPH.

2025 **Dept. of Computer Science, UCPH**, *guest lecturer*, PhD course on Machine Learning for Sciences, "Physics-Informed Machine Learning as a Carbon-Efficient Approach".

2025 **Mathematics A**, *teaching assistant*, high school, Falkonergårdens Gymnasium.

2022-23 **Mathematics for Physicists 2**, *teaching assistant*, BSc course, Niels Bohr Institute, UCPH.

2022-23 **Mathematics for Physicists 1**, *teaching assistant*, BSc course, Niels Bohr Institute, UCPH.

2020-21 **Mechanics and Special Relativity**, *teaching assistant*, BSc course, Niels Bohr Institute, UCPH.

Supervision

2026 **Two BSc students**, *co-supervised with R. Selvan*, "The Environmental Impact of Generative AI: A Life Cycle Assessment as a Basis for Studying Consumer Perception", Dept. of Communication, UCPH.

Selected Talks and Panels

Oct. 2026 **Brandenburgisches Zentrum für Medienwissenschaften, Potsdam, Germany**, invited talk: "Hyper-datafication and its Environmental, Social, and Economic Costs".

Oct. 2026 **Halfspace**, invited talk: "Assessing and Mitigating Sustainable Costs of Frontier AI".

Sep. 2026 **High Energy Physics Software Foundation Seminars**, webinar: "Assessing and Mitigating Sustainable Costs of Frontier AI".

Sep. 2026 **Center for AI, Danish School of Media and Journalism**, webinar: "Bæredygtig AI: Udfordringer og Løsninger".

Jun. 2026 **Folkemødet, Bornholm**, panel discussion: "Kærlighed, Klimakrise og Kunstig Intelligens".

May 2026 **Pioneer Centre for AI, UCPH**, invited talk (workshop on Green AI): "How Hyper-Datafication Impacts the Sustainability Costs in Frontier AI".

May 2026 **National Center for AI in Society, UCPH**, invited talk (workshop on Sustainable AI): "On the Challenges in Assessing the Sustainability of AI" (joint with R. Selvan).

Apr. 2026 **Lund Stem Cell Center, Lund University**, invited talk and panel discussion (workshop on Sustainable AI in Practice): "How to measure and manage energy consumption of AI models".

Academic Service & Organisation

2026 **SAINTS Lab Journal Club**, *coordinator*.

Oct. 2026 **D3A 4.0 Workshop on Resource-Aware and -Constraint AI**, *co-organiser*.

Sep. 2026 **AAAI 2026 Fall Symposium on Parsimonious Scientific Machine Learning**, *programme committee member*.

Sep. 2026 **BSc in Machine Learning, Dept. of Computer Science, UCPH**, *admissions committee member*.

Mar. 2026 **SAINTS'26 Workshop on Sustainable AI for Sciences**, *co-organiser and moderator*.

Outreach

Sep. 2025 **Det Åbne Gymnasium**, workshop: "Energiforbrug og CO_2-aftryk af kunstig intelligens".

May 2025 **Falkonergårdens Gymnasium**, three talks: "Hvordan er det at læse fysik på universitetet?".

2023 **Summer Science Camp**, volunteer: elementary school, Insero.

2021-24 **Women in Physics**, contributing to initiatives promoting gender balance in STEM.

Media

Sep. 2026 **Podcast guest in Verden Forsøgt Forklaret**.
`https://open.spotify.com/episode/7wsAAKJJaPAr6avWYEUfF3?si=dee33aab14d1460b&nd=1&dlsi=1c3cff7934644ef7`

Sep. 2026 **Interview by Uniavisen**.
`https://uniavisen.dk/forsker-ai-bliver-aldrig-en-baeredygtig-teknologi/`

Dec. 2025 **Interview by Women in Physics**.
`https://kvinderifysik.dk/2025/12/15/interview-with-sophia-wilson/`